ABIGAIL DITTMAR

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Education

Florida State University – Tallahassee, FL 2020-present
Ph.D. in Biological Science, Area: Ecology and Evolution
Committee: Nora Underwood (chair), Brian Inouye (chair), Michael Cortez, Alice Winn, Emily Duval, Nicholas Cogan

Cornell University – Ithaca, NY May 2020
B.A. in Biological Sciences, concentration in Ecology and Evolutionary Biology
Magna Cum Laude

Research Experience

Graduate Research Assistant – Underwood Inouye Lab | FSU 2020-present

- Studying how mechanisms of plant resistance to herbivory affect spatial distributions of plants and insects over time using mathematical models.
- Built and analyzed a discrete-time difference equation model of effects of plant induction on a stage-structured insect population.
- Created and analyzed a spatially explicit individual-based model of two plant species interacting over time.
- Designed and analyzed a partial differential equation model of an insect feeding on a plant with induced resistance over time and space.
- Developed a simulation model to explore null models to verify against mechanisms driving variable herbivory.
- Developed a simulation model to generate predictions of how plant traits influence variation in herbivory.
- Participated in field research on plant associational effects and herbivory.
- Designed and conducted lab experiment exploring induction specificity in *Solanum lycopersicum* to different instars of *Manduca sexta* and *Leptinotarsa decemlineata*.

Honors Thesis – Thaler Lab | Cornell University 2019-2020

- Designed project exploring pheromone release in *Podisus maculiventris.*
- Executed project using metal oxide sensing and visual observation to examine pheromone release behavior in *P. maculiventris.*

Research Assistant – Thaler Lab | Cornell University 2017-2020

- Participated in field research on effects of predator semiochemicals on herbivore feeding.

- Performed lab studies on effects of predator semiochemicals on herbivore behavior and plant chemistry.
- Maintained colonies of *Podisus maculiventris* (spined soldier bug), *Leptinotarsa decemlineata* (Colorado potato beetle), and *Trichoplusia ni* (cabbage looper).

Awards and Honors

- FSU Legacy Fellowship ($50,000) 2020-2025
- Dean's List, Cornell University 2016-2019

Grants

FSU Seed Grant 2023-2024
Modeling the role of plant induced defenses in herbivore population growth.
PI Brian Inouye. Coauthor Abigail Dittmar. ($29,928)

Teaching and Advising

Undergraduate Advising

Anna Chilton (2026)

- Advised on an individual research project for the Herbivory Variability Network on null models
- Guided Anna through new skills in coding, probability models, writing, and presenting research

Graduate Teaching Assistant – Florida State University 2020-present

Biological Science II Laboratory (BSC 2011L)
General Ecology (PCB 3043)
Experimental Biology Laboratory, Plant-Insect Interactions (BSC 3402L)
Plants and Society (BOT 4802)
Field Botany (BOT 3143c)
Animal Diversity Laboratory (ZOO 3141L)
Animal Diversity Laboratory, Lead TA (ZOO3141L)

First-Year Student Advisor – Cornell University 2019-2020

- Guided students in building academic schedules and helped them find opportunities on campus.

Arts and Sciences Ambassador – Cornell University 2018-2020

- Led information sessions and met individually with prospective students to provide guidance in the college admissions process.

Presentations

Invited Oral Presentations

Dittmar, A.G., N. Underwood, and B.D. Inouye. (2023, Sept.) Modeling stage-specific effects of induced resistance on herbivore populations. University of Georgia Odum School of Ecology Theory Lunch.

Oral Presentations

Dittmar, A.G., K.C. Abbott, B.D. Elderd, and B.D. Inouye. (2026, Aug). Signal or noise: is variation in herbivory among plants driven by biology or chance? Ecological Society of America annual meeting.

Dittmar, A.G., N. Underwood, and B.D. Inouye. (2023, Aug.). Modeling stage specific effects of induced resistance on herbivore populations. Ecological Society of America annual meeting.

Dittmar, A.G., L.K. M'Gonigle, N. Underwood, and B.D. Inouye. (2022, Aug.). Neighborhood effects and competition alter dispersion and coexistence. Ecological Society of America annual meeting.

Poster Presentations

Dittmar, A.G., N. Underwood, and B.D. Inouye. (2024, June). Modeling stage-specific effects of induced resistance on herbivore populations. Society of Industrial and Applied Mathematics – Life Sciences Meeting.

Publications

Aflitto, N.C., A.G. Dittmar, T.A. Ugine, and J.S. Thaler. 2023. Semiochemical release and ontogenetic changes in a primary scent gland of *Podisus maculiventris*. *Journal of Chemical Ecology* 49:428-436.

Working Papers

Dittmar, A.G., N. Underwood, and B.D. Inouye. Herbivore stage-structure modifies effects of induced resistance on insect population dynamics. *Resubmitted after revision to Journal of Animal Ecology*.

Dittmar, A.G., L.K. M'Gonigle, and B.D. Inouye. Neighborhood effects and competition alter dispersion and coexistence of plant communities. *In prep.*

Dittmar, A.G., et al. Signal or noise: is variation in herbivory among plants driven by biology or chance? *In prep*

Galmán, A., et al. Beyond the mean: how plant growth-defense strategies influence intrapopulation variation in herbivory. *In prep*.

Service

Reviewing Activity for Journals
Journal of Ecology (2026)
Proceedings B (2025)
Forest Ecology and Management (2020)

EERDG Historian 2023-2024

- Served on the executive board for the Florida State University Ecology and Evolution Research Discussion Group.
- Helped to organize events to foster community within the department and among graduate students.
- Ran the group's social media to showcase graduate student research, outreach, and community-building efforts.

Scientific Outreach and Education

K-12 School Visits, Scientists in Every Florida School

- Developed and taught a lesson plan about local Florida ecosystems for middle school aged students at Okaloosa STEMM Academy (Spring 2023, 2024)
- Created and taught a lesson plan about biodiversity and evolution for high school aged students at Florida A&M University Developmental Research School. (Spring 2024, 2025)
- Designed and taught a lesson plan about plants and their adaptations for third grade students at Crawfordville Elementary School. (Fall 2024)
- Developed and taught a lesson plan about food plant adaptations for third graders at a school in Seminole County, FL. (Spring 2022)

Letters to a Pre-Scientist (Fall 2025, Spring 2026)

- Wrote letters to an 8th grade student pen-pal at a title 1 school to increase student engagement with science and make scientists more accessible.

Tallahassee Science Festival (Fall 2021, 2022, 2023, 2025)

- Taught local community about native Florida plants and animals at annual science festival.
- Focused on organismal behavior, physiology, and interactions as well as how people can get involved with citizen science initiatives.

Encouraging Young Engineers and Scientists (2018-2019)

- Designed and led lessons about scientific concepts for groups of 5-20 elementary-aged kids.

Expanding Your Horizons (Spring 2018, 2019)

- Acted as a responsible guardian and mentor for seventh grade girls at an outreach event for middle school girls interested in science and math careers.

Insectapalooza (Fall 2017, 2018)

- Taught visitors about behavior and chemistry of *Manduca sexta* for annual insect fair.